

Detection and characterisation of non-transiting exoplanets with orbital brightness modulation

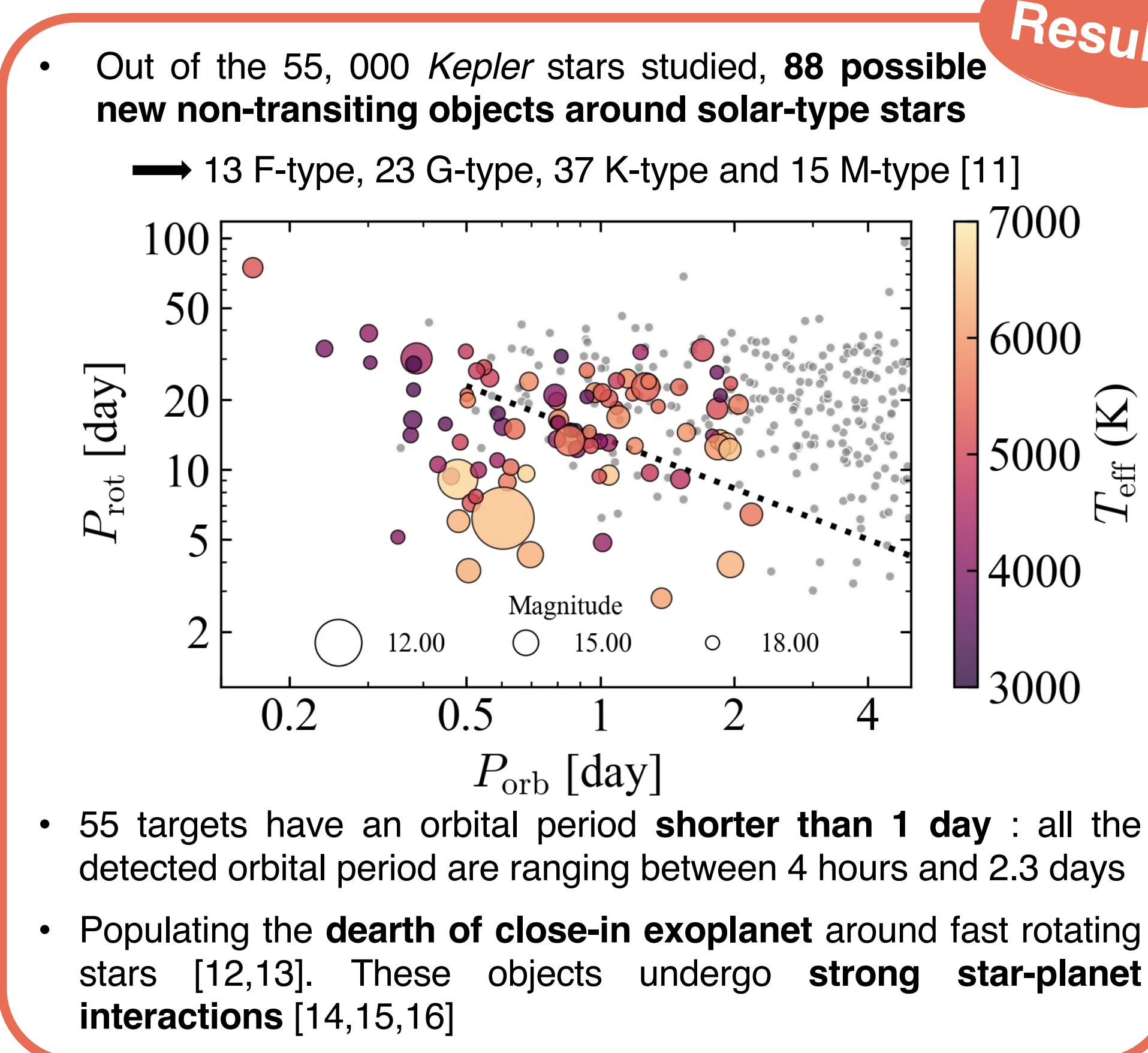
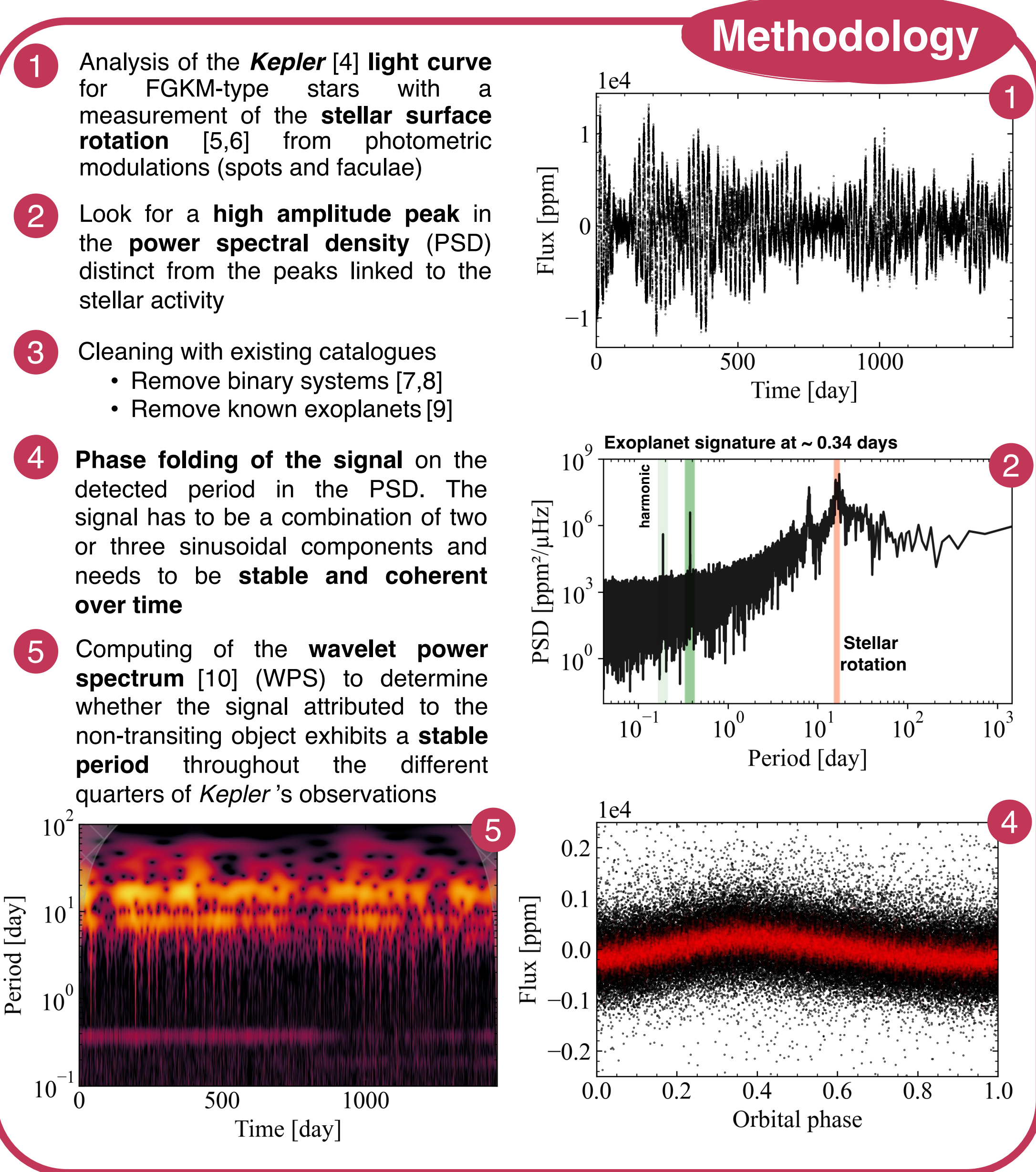
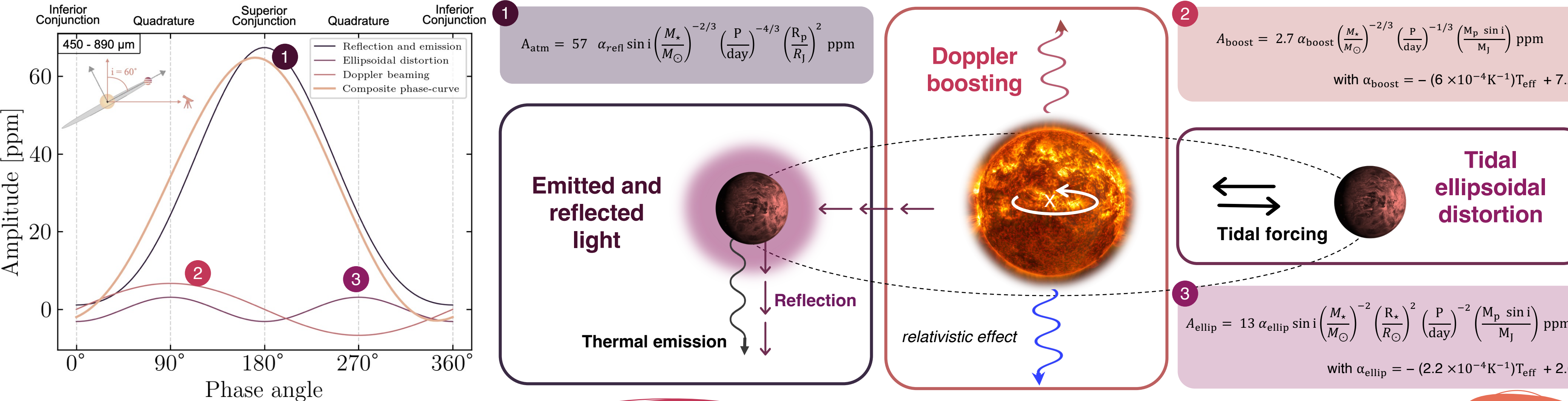
Clémence Gourvès¹, S.N. Breton², A. Dyrek³, R.A. García¹, A.F. Lanza², S. Mathur^{4,5}, J. Morin⁶, C. Moutou⁷, P. Petit⁷, Â.R.G. Santos¹, and A. Strugarek¹

Context

To fully characterise the architecture and diversity of exoplanetary systems, it is essential to expand our focus beyond the geometric limits of the transit method. While the Ariel mission will focus on delivering unprecedented atmospheric data for transiting exoplanets, a comprehensive understanding of atmospheric exoplanet populations would greatly benefit in having more targets, for example by including the **non-transiting exoplanets systems** that currently remain largely unexplored. **Close-in planets produce large orbital phase-curve modulations**, enabling their detection in photometry even without transits. However, out of the 6,107 confirmed exoplanets, only 33 non-transiting planets exhibit orbital periods shorter than 3 days. To identify more close-in non-transiting systems, we use the **orbital brightness modulation** method [1,2].

➤ We identify 88 stars exhibiting a signature consistent with the presence of a close non-transiting substellar companion in the *Kepler* data.

The photometric modulations linked to the presence of a planet around its host star arise from three main physical processes [3]



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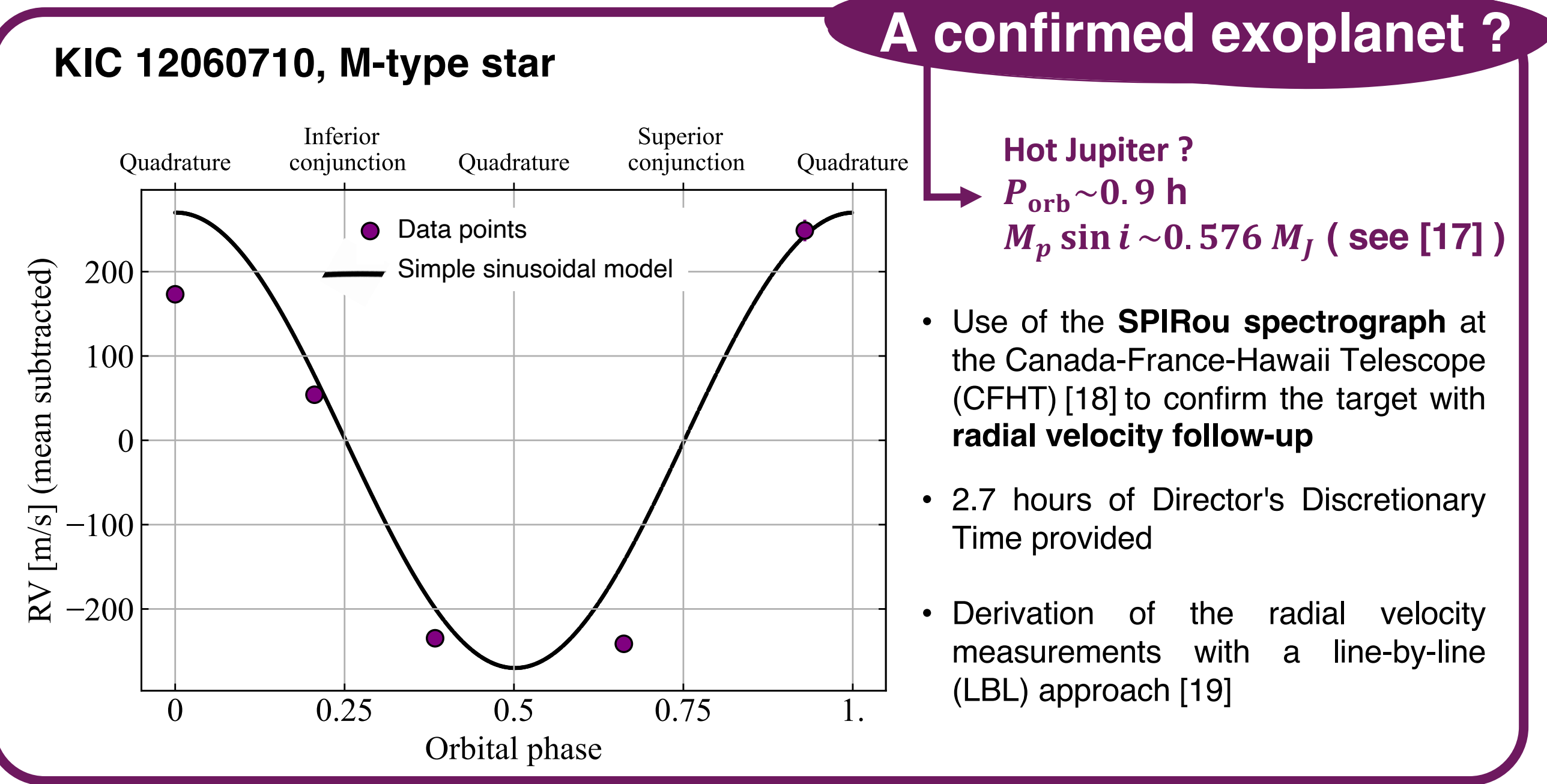
Conclusion

- ✓ Detection in photometry of **88 new non-transiting objects** with very short orbital period
- ✓ Pre-confirmation of the existence of KIC 12060710b through **radial-velocity follow-up** with SPIROU
- ✓ Our targets would constitute promising candidates for **atmospheric characterisation with Ariel** using Mid-infrared excess method [20,21]
- ✓ Use follow-up data to constrain **key planetary parameters** (e.g. M_p , R_p , i , albedo)

Futures perspectives

- Apply our methodology to photometry missions such as **TESS** [22] and **PLATO**
- Upcoming observational campaign of 15 hours to definitely confirm **KIC 12060710b** and **KIC 7847566b**
- An expanded laboratory for studying **star-planet magnetic interactions** [23]

➔ Contact at clemence.gourves@cea.fr



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